

Proportional Reasoning — Full Test

Mixed questions from all three concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes ·
Total: 28 marks. Answers on the separate key.

Objective (1 mark each)**Q1 · MCQ****[1]**

The simplest form of $24 : 18$ is:

- (a) $12 : 9$ (b) $4 : 3$ (c) $3 : 4$ (d) $6 : 5$

Q2 · MCQ**[1]**

If $a : b :: c : d$, then:

- (a) $ac = bd$ (b) $ad = bc$ (c) $ab = cd$ (d) $a + d = b + c$

Q3 · MCQ**[1]**

Which is a true proportion?

- (a) $8 : 3 :: 24 : 6$ (b) $21 : 6 :: 35 : 10$ (c) $7 : 12 :: 12 : 7$ (d) $12 : 18 :: 28 : 12$

Q4 · MCQ**[1]**

₹100 shared in the ratio $3 : 2$ — the larger share is:

- (a) ₹40 (b) ₹60 (c) ₹50 (d) ₹75

Q5 · FILL IN THE BLANK**[1]**

In $6 : 10 :: 18 : x$, the value of x is _____.

Q6 · TRUE / FALSE**[1]**

1 litre = 1000 mL.

True False

Q7 · MCQ

[1]

25 °C in Fahrenheit is:

(a) 45 °F (b) 77 °F (c) 57 °F (d) 98 °F

Q8 · ASSERTION-REASON

[1]

Assertion (A): The ratios 72 : 96 and 3 : 4 are proportional.

Reason (R): Both have the same simplest form, 3 : 4.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

Write 600 : 900 in its simplest form.

Q10

[2]

Find x: 5 : 8 :: x : 24.

Q11

[2]

Divide 42 in the ratio 4 : 3.

Q12

[2]

Convert 25 °C to Fahrenheit.

Long answer (3 marks each)

Q13

[3]

A car travels 90 km in 150 minutes. At the same speed, how far will it travel in 4 hours?

Q14

[3]

A 40 kg mortar mix has sand : cement = 3 : 1. Find the weight of the sand and the cement.

Q15

[3]

Divide ₹360 between two people in the ratio 5 : 4.

Q16

[3]

When Neelima is 3 her mother is 30. Write the ratio of their ages in simplest form, and again when Neelima is 12 (mother 39). Are they the same?

Total: 28 marks · Check your work against the **answer key**.